Jason Guan

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Work Experience

DefTechLink Tampa, FL

Full-Stack Developer Lead *March 2025 – July 2026*

- Led full-stack development of two production B2B SaaS platforms that aggregate and manage global defense contracting opportunities for technology companies, using React, Flask, PostgreSQL, and AWS.
- Built an AWS Lambda and EventBridge pipeline ingesting 1,000+ defense contract notices daily from global government sources, with automated scoring, tagging, and classification.
- Developed OpenAI-powered workflows for opportunity matching, executive report generation, and automated content analysis.
- Integrated Stripe subscription billing, QuickBooks payment reconciliation, AWS S3 storage, and Mailchimp customer engagement workflows.
- Implemented Auth0 authentication, JWT validation, and role-based authorization for Flask APIs and application workflows.
- Designed PostgreSQL schemas on AWS RDS and executed production-safe migrations to preserve data integrity across development and production environments.

U.S. Department of the Air Force Warner Robins, GA

Software Engineer Intern *June 2024 – July 2024*

- Served as backend lead for a management platform centralizing project metrics and sprint tracking across 10+ engineering projects.
- Developed 10+ REST API endpoints using Flask, SQLAlchemy, and PostgreSQL; containerized PostgreSQL with Docker for a platform designed to support 300+ engineers.
- Collaborated across three Agile sprints using Jira and Figma to translate project requirements into application features.
- **U.S. Government Secret Clearance:** Granted in 2024

Education

University of South Florida Tampa, FL

B.S. in Computer Science *Aug 2021 — May 2025*

Certifications: AWS Cloud Practitioner · Issued: October 2026

Skills

Languages: Python, JavaScript, TypeScript, SQL, Java, C++, C, HTML/CSS
Frameworks & Libraries: React, Flask, SQLAlchemy, Node.js, Tailwind CSS, MUI
Cloud & Data: AWS (Lambda, EventBridge, RDS, S3, EC2, CloudWatch, IAM), PostgreSQL, MySQL, SQLite
APIs & Security: REST APIs, Auth0, JWT/OAuth, OpenAI, Stripe, QuickBooks
Testing & Delivery: Vitest, pytest, Playwright, GitHub Actions, Alembic, Docker, Git, Linux

Projects

TaskNook GitHub

React, Flask, SQLite, Alembic, PyInstaller

- Built a desktop productivity app combining task scheduling, recurring routines, and a Pomodoro timer with a customizable isometric workspace, ambient audio, and time-of-day themes.
- Developed an interactive SVG room editor with drag-and-drop furniture placement, depth sorting, and customizable floor plans.
- Implemented database upgrades with automatic SQLite backups and regression tests; configured GitHub Actions for Vitest, pytest, migration checks, and packaged Windows smoke tests.

PokéSQL GitHub

React, TypeScript, SQLite, WebAssembly, Python, Electron

- Built an interactive SQL practice application with a bundled relational dataset of 1,025 Pokémon, a syntax-highlighted editor, schema autocomplete, and eight guided challenges with isolated answer checking.
- Moved SQLite query execution into a Web Worker with cancellation and timeout recovery; persisted database snapshots, query history, and challenge progress locally with IndexedDB.
- Packaged an offline Windows desktop app with Electron and added browser and desktop checks for persistence, transactions, query interruption, and offline reload.

TwitterFeel - Sentiment Analysis GitHub

Python, TensorFlow, Hugging Face, scikit-learn, pandas, NumPy

- Built a machine learning pipeline to predict a user's next-tweet emotion from posting history, using transformer embeddings and TensorFlow LSTM models.
- Evaluated generalization with user-disjoint train/test splits and compared predictions against majority-class and previous-emotion persistence baselines.
- Extended the research workflow with audited data preparation, historical emotion features, shared development folds, per-user metrics, and reproducible run artifacts.

Purrfect Path GitHub

Python, Flask, React, PostgreSQL, AWS

- Built a full-stack course scheduling platform for University of South Florida Computer Science students, supporting academic planning across 28 core and elective courses.
- Implemented workflows for students to create, compare, and share course schedules with friends through a cat-themed interface.

SERAPH – Engineering Project Dashboard GitHub

React, Flask, SQLAlchemy, PostgreSQL, Docker

- Served as backend lead for an engineering management platform centralizing project metrics and sprint tracking across 10+ projects during a U.S. Department of the Air Force internship.
- Developed 10+ REST API endpoints with Flask and SQLAlchemy and containerized PostgreSQL with Docker for a platform designed to support 300+ engineers.

Projects (continued)

Dopamine – Restaurant Ordering Platform GitHub

React, Vite, Flask, SQLAlchemy, SQLite, pytest

- Built a full-stack restaurant ordering application with menu browsing, a persistent cart, pickup and delivery checkout, order tracking, and a kitchen dashboard.
- Implemented server-side price calculation in integer cents, order-line price snapshots, and validated order-status transitions to preserve pricing and order history.
- Added API-contract tests using a disposable database to verify ordering behavior; implemented request rate limiting and key-protected kitchen routes.

Media Grabber GitHub

React, Node.js, Express, yt-dlp, FFmpeg, MUI

- Built a media download application for YouTube, Instagram, and X with format and quality selection, background download jobs, and progress polling.
- Integrated yt-dlp and FFmpeg for video/audio extraction and merging; streamed image conversion through a server proxy and assembled selected photos into ZIP archives in the browser.
- Managed per-job temporary directories with cleanup after downloads and periodic removal of abandoned files; packaged the frontend, API, and native tools as a portable Windows application.

Study with Soobin GitHub

React, TypeScript, Vite, Tailwind CSS, YouTube IFrame API

- Built a browser-based study app combining embedded video, configurable Pomodoro cycles, task planning, and locally persisted focus statistics and preferences.
- Implemented video discovery with search and filters, playback controls, keyboard shortcuts, and saved playback positions; maintained timer and music state while switching tool panels.
- Created a procedural ambient-sound mixer with independently adjustable rain, wind, fireplace, and café channels using the Web Audio API.

Personal Portfolio GitHub

React, Vite, JavaScript, CSS, GSAP, GitHub Actions

- Designed and built a responsive personal portfolio with a custom café-inspired visual system, project galleries, and a YouTube-powered music player.
- Implemented keyboard-accessible modal galleries with focus restoration, reduced-motion support, and responsive navigation; configured automated GitHub Pages deployment with GitHub Actions.

Autonomous Robot GitHub

Arduino, C++, TinkerCAD, Ultrasonic Sensors, 3D Printing

- Prototyped a $30 autonomous robot as an affordable STEM toy for K–12 students, using Arduino, C++, and ultrasonic sensors for obstacle-course navigation.
- Modeled a train-themed enclosure in TinkerCAD, then 3D printed and assembled the robot hardware.